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A comparative study of solvent-assisted pretreatment of biodiesel derived crude glycerol on growth and 1,3-propanediol production from *Citrobacter freundii*

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The rapidly growing biodiesel industry has created a scenario, where it is both important and challenging to deal with the enormous amount of crude glycerol generated as an inherent by-product. With every 100 gallons of biodiesel produced, 5–10 gallons of the crude glycerol is left behind containing several impurities which makes its disposal difficult. The objective of the present investigation was to evaluate the impact of biodiesel-derived crude glycerol upon microbial growth and production of 1,3-propanediol by *Citrobacter freundii*. Five different grades of crude glycerol (obtained from biodiesel preparation using jatropha, soybean, sunflower, rice bran and linseed oils) were used. Crude glycerol caused significant inhibition of microbial growth and subsequently 1,3-propanediol production as compared to pure glycerol. Therefore, a process was developed for the treatment of crude glycerol using solvents before fermentation wherein four different non-polar solvents were examined yielding different grades of pretreated glycerol. Subsequently, the potential toxic effects of pretreated glycerol on the growth and 1,3-propanediol production by *C. freundii* was evaluated. In case of petroleum ether-treated crude glycerol obtained from jatropha & linseed and hexane-treated crude glycerol obtained from rice bran, the yields obtained were comparable to the pure glycerol. Similarly, soybean-derived glycerol gave comparable results after treatment with either hexane or petroleum ether.

Introduction

With rapidly increasing environmental concerns, the stakes are exceedingly high for the search of greener substitutes for various petrochemical products including fuels [1]. In this context, the global biodiesel industry has recently experienced a major surge worldwide and has been on an exponential growth curve over the past several years [2]. European Union biodiesel board reported that the production capacity has currently reached some 22 million tones [3]. Some of the main drivers behind this tremendous growth are reducing dependence on imported oil, environmental-friendly alternative to diesel and reducing greenhouse gas emission. However, there is almost the unanimous opinion among both proponents and skeptics that the potential of the biodiesel requires attention to both safety and economical issues.

The global biodiesel market is estimated to reach 37 billion gallons by 2016 with an average annual growth of 42%, which means about 4 billion gallons of crude glycerol will be produced [4]. Because of this, glycerol stemming from biodiesel production is flooding the market. Finally the euphoria once associated with biodiesel production has given way to disappointment because of lack of economic viability as enormous amount of waste glycerols generated in its preparation [5]. At present, most of it is disposed off in rivers or simply incinerated, making biodiesel a grey fuel rather a green fuel. Although glycerol valorization has much to offer in reducing the cost of the overall biodiesel production process in spite of the continuous efforts the realization of this vision is difficult to achieve. This is because of the several impurities present in the crude glycerol such as residual methanol, NaOH, carry-over fat/oil, some esters, and low amounts of sulfur compounds, proteins, and minerals [6]. This has not only been a serious handicap in itself for being cost effective, but also served as

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a major drag in the development of new alternative fuels that can be added to the limited armory of green fuels.

Thus, valorization of this low value crude glycerol into value-added products via microbiological or chemical routes currently attracts much and continuously increasing interest [7,8]. In this context, the well-known biotechnological conversion of glycerol into 1,3-propanediol provides a greener solution to this darker side of biodiesel [1]. The first description of the microbial production of 1,3-propanediol by August Freund in 1881 illustrates that it is a product of glycerol fermentation [9]. However, a lot of knowledge about this peculiar biochemical conversion has been gained since then, but the significance of 1,3-propanediol was fully recognized when its importance as a starting material, for a new generation polymer, was established. Because of the hydroxyl group at terminal positions, 1,3-propanediol finds application in polymer industry [10]. Now, great efforts are being exerted globally for its production [11–13] using microbial route and its purification from fermentation broth [14]. The best-known producers investigated so far belong to Clostridiaceae (*Clostridium butyricum*) [15,16] and Enterobacteriaceae (*Klebsiella pneumoniae* [17], and *Citrobacter freundii* [18]).

However, there are only few reports evaluating the feasibility of crude glycerol as a substrate for the conversion to 1,3-propanediol [19–21]. Petitdemange *et al.* [19] observed that industrial glycerol obtained through the transesterification process using rapeseed oil did not support growth of several strains of *C. butyricum* obtained from bacterial culture collections. Mu *et al.* [22] also examined the combined process of biodiesel production (by lipase) and 1,3-propanediol production in which glycerol was passed through the membrane for 1,3-propanediol production. Chatzifragkou *et al.* [23] while evaluating the effect of biodiesel-derived waste glycerol impurities on biomass and 1,3-propanediol production from *C. butyricum* found that oleic acid was the restraining factor for microbial growth. Rehman *et al.* [24] reported an efficient pretreatment process for crude glycerol by solvent washing. However, they used only sunflower based biodiesel-derived waste glycerol.

In our previous investigation, we attempted solvent washing (hexane) of crude glycerol obtained from biodiesel production using six different oils [1]. It was found that the variation in 1,3-propanediol production was in fact due to the differences of impurities in crude glycerol. Therefore, a single method cannot be fully applicable for the treatment of crude glycerol obtained from different oil sources, because the fatty acid composition differs in various oil which is the main inhibitory factor in 1,3-propanediol production. In this context, Moon *et al.* [25] reported that the inhibitory effect of raw glycerol from soybean-oil-based biodiesel production was less than that of raw glycerol from waste vegetable-oil-based biodiesel production because of the presence of different concentrations of various impurities. Rehman *et al.* [24] reported that the cell viability of *C. butyricum* was significantly affected by the unsaturated fatty acids such as linoleic and oleic acid present in the crude glycerol obtained from sunflower oil.

In this context, in the present investigation, 5 different grades of crude glycerol were obtained from biodiesel production processes using different oils. Pretreatment of crude glycerol was performed by solvent washing where four different solvents were used separately to remove impurities. Subsequently all these grades of crude and pretreated glycerol were used as a substrate to produce 1,3-propanediol in batch fermentation. Growth inhibition of *C. freundii* and 1,3-propanediol production using different grades of glycerols was evaluated and compared with the commercial grade pure glycerol.

Materials and methods

Materials

1,3-Propanediol and glycerol were purchased from Sigma chemicals (St. Louis, USA). Other chemicals used were purchased from commercial sources (Hi-Media, Qualigenes and Sisco Research Laboratories Ltd., India) at the highest available purity.

Microorganism and growth conditions

In the present investigation, 1,3-propanediol-producing *C. freundii* was used. This organism was grown in 500 mL anaerobic bottles containing 200 mL of the medium which consisted of (g/L): glycerol, 50; glucose, 1.5; tryptone, 25.0; $K_2HPO_4{\cdot}3H_2O$, 5.0; KH_2PO_4, 3.48; $(NH_4)_2SO_4$, 2.0; $MgSO_4{\cdot}7H_2O$, 0.4; $CaCl_2{\cdot}2H_2O$, 0.2; $MgCO_3$, 150 mM; $CoCl_2{\cdot}6H_2O$, 4.0 mg, at pH 7.0. Glycerol used in the above medium were obtained from three different sources, that is, pure commercial glycerol (99%, w/w); crude glycerol obtained from biodiesel preparation; pretreated glycerol after solvent washing. To secure anaerobiosis, infusion of CO_2 gas was performed. The medium was sealed with butyl rubber bungs with CO_2 headspace and sterilized for 15 min at 121°C. To remove traces of dissolved oxygen, $Na_2S{\cdot}9H_2O$ (0.02%, v/v) was added before inoculation. The reduced medium was then inoculated with 4.0% (v/v) inoculum (0.7–0.8 O.D. at 660 nm) and incubated at 30 $\pm$ 1°C and 150 rpm.

Inoculum was raised at 37 $\pm$ 1°C in 250 mL shake flasks containing 50 mL of preculture medium at 150 rpm under aerobic conditions. Preculture medium contained (g/L): beef extract, 1.0; yeast extract, 2.0; peptone, 5.0; NaCl, 5.0 at pH 7.0.

Biodiesel synthesis

Transesterification of different oils (jatropha, soybean, sunflower, rice bran and linseed) was carried out with pure methanol in a molar ratio (oil:methanol) of 1:4 using 0.5% (w/v) NaOH as a catalyst. First, NaOH was added to methanol, stirred until the catalyst was completely dissolved, then mixed with 1 L of oil and incubated at 45°C at 200 rpm for 12 h under shake flask condition. Following completion of the reaction, the product was kept in a separatory funnel to allow the separation of two phases. Crude glycerol was obtained directly from the bottom phase without any purification and used for further investigation. The glycerol concentration in the bottom phase obtained after biodiesel preparation was 500, 504, 892, 882 and 640 g/L from jatropha, soybean, sunflower, rice bran and linseed oils, respectively.

Solvent washing

Crude glycerol obtained from biodiesel preparation was washed with non-polar solvents (hexane, heptane, octane and petroleum ether) in 1:1 volume ratio at 200 rpm and room temperature for 3 h in separate sets (Fig. 1). After mixing it well, it was centrifuged at 3000 $\times$ g for 2 min to separate into two distinct phases, that is, the upper phase of solvent and the lower aqueous phase of glycerol containing residual impurities. The lower phase was subjected to washing again with the same solvent. The procedure was repeated three times to remove impurities and finally solvent-washed pretreated glycerol was obtained.

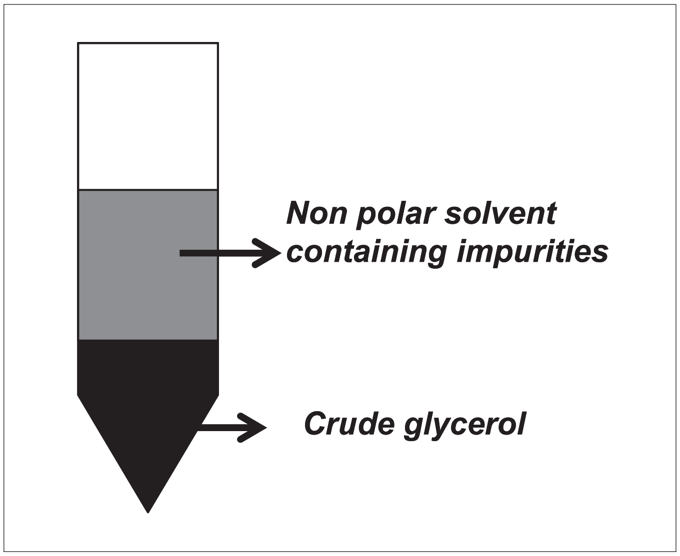

FIGURE 1
Treatment of crude glycerol by solvent washing.

Growth inhibition of C. freundii *by crude and solvent-washed glycerol*

Inhibition due to impurities in crude and pretreated glycerol was studied by determining percent inhibition using test tubes containing 9 mL of culture medium with different grades of glycerols. The medium was inoculated with 1 mL of inoculum and incubated at 30°C. The optical density at 660 nm was measured at 0 and 6 h in each tube. The percent inhibition was determined from the following equation:

$$\text{Growth inhibition}\,(\%) = 100 \times \left(1 - \frac{(OD_{E(t=6h)} - OD_{E(t=0)})}{(OD_{C(t=6h)} - OD_{C(t=0)})}\right)$$

where $OD_{E(t=6h)} - OD_{E(t=0)}$ is increase of OD_{660} in the presence of crude or pretreated glycerol and $OD_{C(t=6h)} - OD_{C(t=0)}$ is increase of OD_{660} in the presence of pure glycerol.

Analytical procedures

Estimation of biomass

After the desired incubation period, the culture was centrifuged at 10,000 × *g* for 15 min (Sigma centrifuge). Cell pellet was used for growth estimation. Here, pellet was washed with 2 mL of distilled water and again centrifuged at 10,000 × *g* for 10 min. The process was repeated three times to remove all media components from the cell surface. Now, the pellet was re-suspended in double distilled water up to the appropriate dilution and the absorbance was read at 660 nm against control using spectrophotometer (UV-Visible 1700, Shimadzu Corp., Kyoto, Japan).

Estimation of fermentation by-products

After the desired incubation period, the culture was centrifuged at 10,000 × *g* for 15 min (Sigma centrifuge). The supernatant thus obtained was filtered through syringe filters (pore size 0.45 μm, Mdi filters, India). Filtrates obtained were analyzed for the concentration of glycerol and 1,3-propanediol using high-performance liquid chromatography system (Shimadzu 10AVP, Shimadzu Corp., Kyoto, Japan) equipped with Aminex HPX-87H column (300 mm × 7.8 mm) (Bio-Rad, Palo Alto, CA, USA) and RID-10A refractive index detector. The working conditions were: 5 mM H_2SO_4 as a mobile phase with a flow rate of 0.6 mL/min at 60°C as the working temperature.

Estimation of free fatty acids and fatty acid methyl esters

Free fatty acids and fatty acid methyl esters were analyzed by a gas chromatography system (GC-2014, Shimadzu Corp., Kyoto, Japan) equipped with a capillary column (Stabilwax-DA, 30 m × 0.32 mm, 0.25 μm film thickness) and flame ionizing detector (FID). Helium gas was used as the carrier gas at a flow rate of 2.87 mL/min. Injection was done in spilt mode (1/5) and the temperatures of injector, detector and oven were at 250, 260 and 160°C, respectively. Samples (1 μL) were injected and then the oven was heated at 4°C/min to 250°C (holding for 10 min). The samples were prepared by filtration using nylon syringe filters (pore size 0.20 μm, Mdi filters, India).

Results and discussion

Efficiency of solvent washing for pretreatment of crude glycerol

Five types of crude glycerol samples obtained from five different oil-based biodiesel syntheses were used in the present investigation. These crude glycerol samples having differences in their purity were initially separated from the biodiesel layer using separatory funnel and were subsequently washed with non-polar solvents (hexane, heptane, octane and petroleum ether) in 1:1 volume ratio in separate sets. Similar washing procedure was repeated three times with the same solvent and finally solvent-washed with pretreated glycerol was obtained. Rehman *et al.* [24] were first to report a pretreatment process of crude glycerol, where sunflower oil based biodiesel-derived glycerol was treated with hexane and hexanol. In our earlier investigation, six different crude glycerol samples obtained from six different oil-based biodiesel syntheses were pretreated with hexane. Subsequently, results of production of 1,3-propanediol and growth inhibition percentage of pure, crude and pretreated glycerol were compared [1]. Two important observations were made from the previous investigation; first, pretreatment of crude glycerol with hexane effectively removed impurities resulting in the 1,3-propanediol production comparable with the pure glycerol. Second, glycerol obtained from the different sources did not give similar results. This was potentially because oils differ in their fatty acids compositions and thereby resulting in different fatty acid impurities in crude glycerol. These findings raised question about whether same process is effective for different grades of glycerol obtained from different oil based biodiesel preparations or not? As the fatty acid content vary in different oils, it is important to investigate different oils commonly used for biodiesel production. Similarly, different solvents should also be investigated to determine the effect of solvents on removal of impurities.

Therefore, in the present investigation, an experiment was designed where five different crude glycerol obtained from five different oil based biodiesel syntheses were washed with four non-polar solvents three times in different sets. Very negligible amount of glycerol was also lost at each step of washing. Figure 2 presents the efficiency of non-polar solvents in removing free fatty acids (FFAs) and fatty acids methyl esters (FAMEs) from different crude glycerols. Glycerol obtained from jatropha oil transesterification was found to be the most impure source which only after three subsequent washing finally resulted in substantial removal of FAMEs and FFAs. In this case, the impurity mainly consisted of

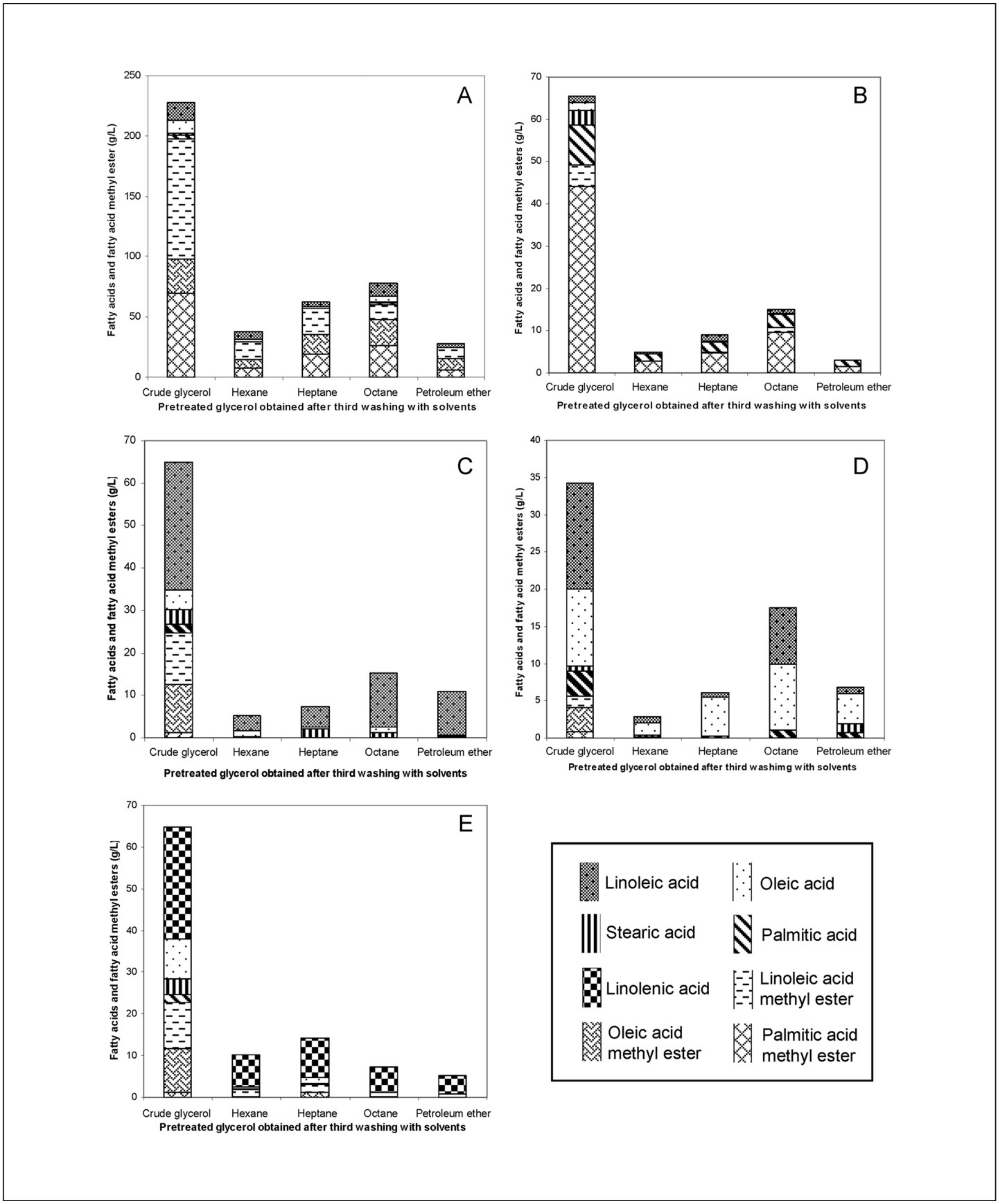

FIGURE 2

Removal of free fatty acids and fatty acid methyl esters by solvent treatment from crude glycerol obtained from **(a)** jatropha, **(b)** soybean, **(c)** sunflower, **(d)** rice bran and **(e)** linseed based biodiesel preparation.

linoleic acid methyl ester (100 g/L) which was successfully reduced by petroleum ether and hexane up to 9.0% and 10.9%, respectively. Another major component palmitic acid methyl ester (69.5 g/L) was reduced by petroleum ether up to 9.2% followed by hexane with 12.5%. Here, the concentration of oleic acid in crude glycerol was 11.1 g/L, which was completely reduced to zero by petroleum ether. The main components of soybean oil based glycerol were palmitic acid methyl ester (44 g/L) and palmitic acid (9.45 g/L) which were efficiently reduced by petroleum ether up to 1.6 g/L and 1.42 g/L, followed by hexane up to 2.7 g/L and 1.82 g/L, respectively.

Following these results, sunflower oil which proved to be the good source of glycerol (892 g/L of glycerol in the bottom phase) after transesterification contained impurities mainly composed of linoleic acid which was approximately 50% of the total impurities. Linoleic acid (30.2 g/L) was reduced to 12.2% and 15% of the initial amount using hexane and heptane, respectively. Other impurities were reduced approximately to 4.0–9.1% using any of the solvents tested. Saturated FFAs do not usually interfere with bacterial growth; however, unsaturated FFAs (particularly oleic acid, linoleic acid and linolenic acid) have profound influence on the viability of bacterial cells [26]. A study by Chatzifraghou *et al.* [23] evaluated the effect of oleic acid upon microbial growth and 1,3-propanediol production. They reported that 2.0% of oleic acid completely restrains microbial growth because of the presence of double bond. In the present investigation, glycerol obtained from rice bran oil based biodiesel preparation had the least minimum impurities mainly composed of linoleic (14.29 g/L) and oleic acid (10.26 g/L) and both were effectively reduced to 6.41% and 15.1%, respectively by hexane. Linseed oil contained linolenic acid (26.7 g/L) and oleic acid (9.7 g/L) as the main impurity. Linolenic acid and oleic acid were successfully reduced by petroleum ether up to 16.1% and 8.2%, respectively; however, by hexane it reduced to 29.2% and 13.4%, respectively. None of the crude glycerol formed gel when mixed with solvents for pretreatment as was observed by Moon *et al.* [25].

Growth inhibition of C. freundii *by crude and solvent-washed pretreated glycerol*

Even in the past, the low-price crude glycerol obtained from various industrial processes were evaluated for the bioconversion to 1,3-propanediol by different workers [19–21,24,27–29]. Petitdemange *et al.* [19] observed that industrial glycerol obtained through the transesterification process using rapeseed oil did not support the growth of several strains of *C. butyricum* obtained from bacterial culture collections. However, they reported that among the ten strains isolated from mud samples, *C. butyricum* E5 was resistant to high concentration (108 g/L) of industrial glycerol. Gonzalez-Pajuelo *et al.* [27] reported that crude glycerol can induce a growth inhibition of 86% on *C. butyricum* VPI 3266, when grown in the fermentation medium with initial concentration of 100 g/L. Papanikolaou *et al.* [29] demonstrated that the effect of crude glycerol (from biodiesel production) on the growth of *C. butyricum* strain F2b was minimal and it also did not interfere with 1,3-propanediol production. However, in the same study, it was reported that during continuous operations at various inlet of crude glycerol concentrations biomass was decreased with the increase in the crude glycerol inlet. In yet another study Papanikolaou *et al.* [30] examined the crude glycerol which was obtained from alkali-catalyzed methanolysis and lipase-catalyzed methanolysis of soybean oil. No adverse effect on microbial growth and metabolism of *C. butyricum* was observed, when the effect of impurities of the industrial feedstock was investigated. There is good number of study reported on *C. butyricum*; however, in comparison there are only few studies which have been done on other potent strains such as *K. pneumoniae* and *C. freundii*.

In the present investigation, the pretreated crude glycerol samples were diluted with distilled water in 1:1 volume ratio for two reasons. Firstly, it decreases the viscosity of the sample. Secondly, at this point, white-oily precipitates were formed potentially because of the potassium and sodium salts of free fatty acids which were removed by filtration through Whatman filter paper No. 1. A similar observation was reported by Celik *et al.* [6] while working with crude glycerol derived from canola and sunflower oil based biodiesel.

In the present investigation, the growth inhibition percentage of *C. freundii* using crude glycerols at 50 g/L was very profound (jatropha, 45.36 ± 1.56%; soybean, 32.41 ± 1.72%; sunflower, 31.25 ± 1.52%; rice bran, 29.25 ± 1.58%; linseed, 32.56 ± 1.48%). Here, growth inhibition percentage has been calculated against the control value, that is, growth of the culture in the medium containing pure glycerol. After pretreatment of crude glycerol samples using solvents, the results of solvent-washed pretreated glycerol were almost similar to the results obtained from pure glycerol as presented in Fig. 3. Petroleum ether and hexane washed crude glycerol obtained from jatropha oil transesterification showed only 4.3 ± 0.22% and 4.7 ± 0.12% of growth inhibition because, both the solvents can effectively remove saturated and unsaturated FFAs and FAMEs. Similar finding was observed with crude glycerol obtained from soybean, rice bran and linseed. However, a comparative analysis revealed that petroleum ether was considered better choice than hexane for soybean (2.37 ± 0.42%) and linseed oil (1.93 ± 0.28%) derived crude glycerol. On the contrary, sunflower oil based glycerol, which contained impurities composed of linoleic acid approximately 50% of the total amount, was successfully washed with heptane resulting in 4.12 ± 0.68% growth inhibition. Whereas, heptane was not considered as the best extractant for any other glycerol tested. In our previous investigation [1], only hexane

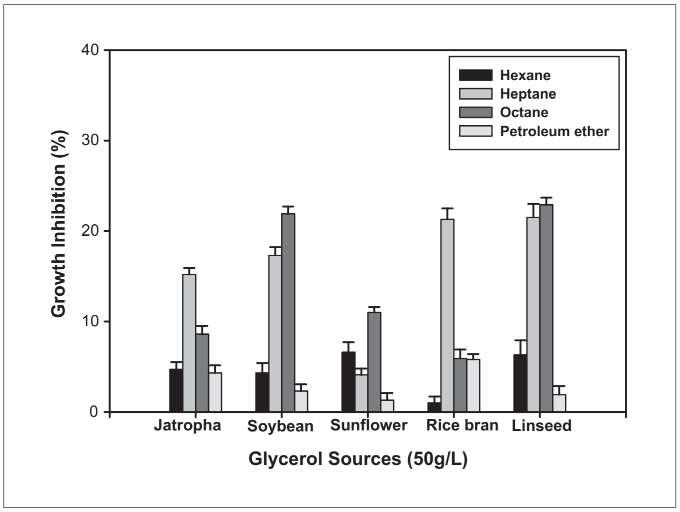

FIGURE 3

Cell growth inhibition of *Citrobacter freundii* in the presence of pretreated glycerol at a concentration of 50 g/L.

was evaluated for pretreatment. In the present investigation, except, for rice bran oil based glycerol, octane washing was found to be not good enough to become a potential extractant. Rehman *et al.* [24] reported that the production of 1,3-propanediol by *C. butyricum* using crude glycerol (from sunflower oil based biodiesel) as a sole carbon and energy source was feasible only when it was washed with different solvents to remove impurities. The inhibitory effects were trounced by washing of crude glycerol with the solvents. In our investigation, Petroleum ether is the best solvent for jatropha, soybean, sunflower and linseed oil based glycerol. Hexane is the best solvent for rice bran oil based glycerol. Reduction in the growth inhibition confirms that the FAs and FAMEs do interfere with the cell growth. Growth inhibition of unsaturated fatty acid on susceptible bacteria is mainly attributed to their physiochemical properties [23].

Comparison of production of 1,3-propanediol and other byproducts using pure, crude and solvent-washed pretreated glycerol by C. freundii

The production of 1,3-propanediol, lactic acid and acetic acid from *C. freundii* using crude and pretreated glycerol derived from the biodiesel preparation is presented in Table 1. In the present investigation, it is clearly evident that crude glycerol obtained from different sources if used as it is, significantly inhibits 1,3-propanediol production. It is also indicated that not all solvents are suitable for pretreatment process. Results showed that *C. freundii* is able to produce 1,3-propanediol from crude glycerol if washed with particular solvent to remove impurities. In the literature, a majority of the reported studies deal with the production of 1,3-propanediol from pure glycerol and there are only few reports on utilization of crude glycerol for 1,3-propanediol production. Barbirato *et al.* [20] demonstrated the production of 1,3-propanediol from *C. butyricum*, *K. pneumoniae, C. freundii,* and *E. agglomerans* using glycerol from stillage (obtained from wine distillation) and waste from ester industry. With waste from stillage (characterized by a low glycerol concentration and a high non-glycerol content), the fermentation was operated without nutrient supply resulted in 0.50 mol/mol of 1,3-propanediol. However, with waste from the ester industry (high glycerol content), a higher yield of 1,3-propanediol concentration (0.69 mol/mol) was obtained in the fermentation, but it required the addition of nutrients. Batch fermentation and continuous fermentation by a strain of *C. butyricum* were carried out on industrial glycerol from the biodiesel production process by Papanikolaou

TABLE 1

Concentration of glycerol, 1,3-propanediol and by-products remained/produced during the fermentation from *Citrobacter freundii* using pure, crude and pretreated glycerol derived from the biodiesel preparation

Source of glycerol	Pretreatment with solvent	1,3-propanediol (g/L)	Residual glycerol (g/L)	Lactic acid (g/L)	Acetic acid (g/L)
Pure glycerol	–	25.36 ± 0.75	12.35 ± 0.34	4.63 ± 0.14	5.47 ± 0.15
Jatropha based biodiesel preparation	Crude	1.40 ± 0.04	41.23 ± 1.24	1.84 ± 0.46	2.32 ± 0.09
	Hexane	23.31 ± 0.69	13.25 ± 0.38	4.45 ± 0.16	6.32 ± 0.17
	Heptane	6.30 ± 0.18	32.56 ± 0.98	3.32 ± 0.17	2.92 ± 0.08
	Octane	12.75 ± 0.46	25.64 ± 0.78	4.43 ± 0.15	4.10 ± 0.16
	Petroleum ether	23.40 ± 0.87	14.58 ± 0.46	4.28 ± 0.26	5.42 ± 0.14
Soybean based biodiesel preparation	Crude	4.65 ± 0.18	34.40 ± 1.09	2.25 ± 0.05	3.54 ± 0.06
	Hexane	23.04 ± 0.64	14.56 ± 0.84	3.38 ± 0.09	5.52 ± 0.16
	Heptane	7.22 ± 0.25	31.58 ± 0.74	2.37 ± 0.06	5.34 ± 0.12
	Octane	5.71 ± 0.18	35.89 ± 0.68	3.31 ± 0.11	3.47 ± 0.08
	Petroleum ether	21.42 ± 0.79	16.41 ± 0.63	5.44 ± 0.15	3.52 ± 0.07
Sunflower based biodiesel preparation	Crude	3.14 ± 0.08	33.67 ± 0.17	2.95 ± 0.04	3.75 ± 0.07
	Hexane	17.98 ± 0.47	19.58 ± 0.54	4.42 ± 0.05	5.39 ± 0.06
	Heptane	24.78 ± 0.79	12.71 ± 0.34	3.39 ± 0.06	4.59 ± 0.08
	Octane	11.52 ± 0.32	24.67 ± 0.56	1.35 ± 0.03	3.38 ± 0.06
	Petroleum ether	25.63 ± 0.86	8.52 ± 0.27	6.32 ± 0.16	4.35 ± 0.12
Rice bran based biodiesel preparation	Crude	4.69 ± 0.05	33.81 ± 0.89	3.14 ± 0.08	2.33 ± 0.06
	Hexane	23.74 ± 0.84	14.36 ± 0.19	4.42 ± 0.16	5.17 ± 0.16
	Heptane	6.99 ± 0.17	31.62 ± 0.75	3.28 ± 0.09	3.45 ± 0.09
	Octane	23.96 ± 0.67	12.87 ± 0.31	4.37 ± 0.15	6.11 ± 0.19
	Petroleum ether	23.94 ± 0.83	11.65 ± 0.15	3.34 ± 0.10	6.15 ± 0.20
Linseed based biodiesel preparation	Crude	4.87 ± 0.12	32.37 ± 0.82	3.32 ± 0.11	2.43 ± 0.12
	Hexane	16.25 ± 0.56	19.85 ± 0.71	2.21 ± 0.06	5.51 ± 0.14
	Heptane	10.25 ± 0.32	26.34 ± 0.81	3.31 ± 0.08	4.42 ± 0.14
	Octane	6.23 ± 0.15	33.64 ± 0.96	1.25 ± 0.05	3.37 ± 0.08
	Petroleum ether	22.41 ± 0.64	14.39 ± 0.96	5.54 ± 0.15	6.32 ± 0.16

et al. [16], for both types of cultures, the conversion yield obtained was around 0.55 g of 1,3-propanediol formed per 1 g of glycerol consumed whereas the highest 1,3-propanediol concentration was achieved during the single-stage continuous cultures which was 48 g/L. Similarly, Papanikolaou and Aggelis [21] evaluated crude glycerol obtained from biodiesel synthesis for the production of 1,3-propanediol, found it to be similar or even better when compared with the commercial glycerol. Similar investigations had been done for the most potent microorganism *K. pneumoniae* [31]. They demonstrated the microbial production of 1,3-propanediol by *K. pneumoniae* by using crude glycerol as the sole carbon source. From our investigation, it is clearly evident that hexane and petroleum ether washed glycerol gave comparable results with pure glycerol. Petroleum ether washed glycerol obtained from jatropha, soybean, sunflower, rice bran and linseed yielded 23.40 ± 0.87, 21.42 ± 0.79, 25.63 ± 0.86, 23.94 ± 0.83 and 22.41 ± 0.64 g/L of 1,3-propanediol. Similarly, hexane washed glycerol obtained from jatropha, soybean and rice bran based biodiesel preparation yielded 23.31 ± 0.69, 23.04 ± 0.64 and 23.74 ± 0.84 g/L 1,3-propanediol which is comparable to the pure glycerol (25.36 ± 0.75 g/L). Similarly, production of lactic acid and acetic acid is also significantly enhanced using solvent-washed crude glycerol.

Conclusions

It can be concluded that *C. freundii* evaluated in the present investigation was not resistant to the various impurities in crude glycerol derived from biodiesel preparation using five different oils. Pretreatment of crude glycerol has proved to be an economical and easy method to remove FFAs and FAMEs which are the most potent growth inhibitors. In addition, these solvents can be easily recovered and reused for the subsequent batches. Furthermore, it was observed that growth was mainly inhibited because of the high concentrations of FFAs and FAMEs. This is proven by the fact that by just removing FFAs and FAMEs, not only the growth inhibition decreased but also 1,3-propanediol production increased to the level which was comparable to the pure glycerol.

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